

Entity Relationship Diagram (ERD) Assignment

1. Entities and Attributes

Each entity below lists its attributes, marking primary keys (PK), foreign keys (FK) and unique keys (UK).

1.1 Customer

Stores registered customers. A customer must register before placing an order.

Entity	Key	Attribute	Notes
Customer	PK	CustomerID	Unique identifier
Customer		FullName	Customer's full name
Customer	UK	Email	Must be unique
Customer		PhoneNumber	
Customer		DateOfRegistration	

1.2 Address

Stores delivery addresses. A customer can have multiple delivery addresses.

Entity	Key	Attribute	Notes
Address	PK	AddressID	Unique identifier
Address	FK	CustomerID	References Customer
Address		AddressLine1	
Address		City	
Address		PostalCode	
Address		Province	

1.3 Category

Product categories (e.g. Electronics, Clothing, Home Appliances).

Entity	Key	Attribute	Notes
Category	PK	CategoryID	Unique identifier
Category		CategoryName	

1.4 Supplier

Companies that supply products to WitleShop.

Entity	Key	Attribute	Notes
Supplier	PK	SupplierID	Unique identifier
Supplier		SupplierName	
Supplier		ContactEmail	
Supplier		ContactPhone	

1.5 Product

Products sold on the platform. Each product belongs to one category and one supplier.

Entity	Key	Attribute	Notes
Product	PK	ProductID	Unique identifier
Product		ProductName	
Product		Description	
Product		Price	
Product		StockQuantity	
Product	FK	CategoryID	References Category
Product	FK	SupplierID	References Supplier

1.6 Order

An order placed by a customer. An order belongs to one customer and can contain multiple products.

Entity	Key	Attribute	Notes
Order	PK	OrderID	Unique identifier
Order	FK	CustomerID	References Customer
Order		OrderDate	
Order		OrderStatus	Pending / Shipped / Delivered / Cancelled
Order		TotalAmount	

1.7 Order_Item

Resolves the many-to-many relationship between Order and Product:

A order can have multiple products, and a product can appear in multiple orders.

Entity	Key	Attribute	Notes
Order_Item	PK	OrderItemID	Unique identifier
Order_Item	FK	OrderID	References Order
Order_Item	FK	ProductID	References Product
Order_Item		Quantity	
Order_Item		UnitPrice	Price at time of order

1.8 Payment

Each order must have exactly one payment record.

Entity	Key	Attribute	Notes
Payment	PK	PaymentID	Unique identifier
Payment	FK / UK	OrderID	References Order — unique, one payment per order
Payment		PaymentDate	
Payment		PaymentMethod	Card / EFT / PayFast
Payment		PaymentStatus	
Payment		AmountPaid	

1.9 Delivery

Each order must have exactly one delivery record linked to one of the customer's registered addresses.

Entity	Key	Attribute	Notes
Delivery	PK	DeliveryID	Unique identifier
Delivery	FK / UK	OrderID	References Order — unique, one delivery per order
Delivery	FK	AddressID	References Address
Delivery		DeliveryDate	
Delivery		DeliveryStatus	
Delivery		CourierName	
Delivery		TrackingNumber	

2. Relationships and Cardinality

The table below summarises every relationship between entities, its cardinality, and the business rule it enforces.

Relationship	Cardinality	Business Rule
Customer — Address	1 : M	A customer can have multiple delivery addresses; each address belongs to one customer.
Customer — Order	1 : M	A customer can place many orders; each order belongs to one customer.
Category — Product	1 : M	A category can have many products; each product belongs to one category.
Supplier — Product	1 : M	A supplier can supply many products; each product is supplied by one supplier.
Order — Order_Item	1 : M	An order can have multiple order items (line items).
Product — Order_Item	1 : M	A product can appear as an order item in many orders. (Order_Item resolves the underlying M:N between Order and Product.)
Order — Payment	1 : 1	Each order has exactly one payment record, and each payment belongs to exactly one order.
Order — Delivery	1 : 1	Each order has exactly one delivery record, and each delivery belongs to exactly one order.
Address — Delivery	1 : M	A delivery is linked to one registered address; an address can be used for multiple deliveries over time.

Note on Order–Product

The business requirement states an order can contain multiple products, and a product can appear in multiple orders, a many-to-many (M:N) relationship. This is resolved in the relational model using the Order_Item associative entity, which breaks the M:N relationship into two 1:M relationships (Order >> Order_Item and Product >> Order_Item).

3. Entity Relationship Diagram

The diagram below shows all entities, their attributes (with PK, FK and UK indicated), and the relationships with cardinality between them.

